Catherine Zhang

425-504-9803 | cat.zhang2006@gmail.com | linkedin.com/in/catherinezhangg/

Education

University of Washington Seattle, WA
Bachelor of Science in Computer Science, GPA: 3.66/4.0 *Sep. 2024 – June 2028 (Expected)*

- *Coursework: Artificial Intelligence, Computer Graphics, Computer Vision, Data Structures and Algorithms, Hardware/Software Interface, Computational Probability*

Experience

Gameplay Programming Intern June 2026 – October 2026
Aggro Crab | *C#, Unity, Blender* *Seattle, WA*

- Sole programming intern on an unannounced title. Designed a centralized entity manager that distributes visibility checks across staggered coroutine cycles, amortizing per-entity work over successive frames, allowing for a **25%** increase in framerate.
- Developed a custom Unity editor tool for procedurally placing prefabs along a spline, sampling positions along the curve at configurable density with surface-snapping raycasts and gizmo previews, replacing manual hand-placement.
- Animated environmental assets through Blender shape keys, Shader Graph vertex-offset shaders, and rigged Unity Animator clips, and added C# interaction logic and custom particle effects.
- Playtested and reviewed builds for one of the largest PEAK updates, a Unity title with **10M+** copies sold total. Concurrent player count reached **top 15** on Steam after this update.

Undergraduate Research Assistant August 2025 – December 2025
The Machine Agency at UW | *JavaScript, G-Code, 3D Printing* *Seattle, WA*

- Applied p5.fab across 3D print workflows for direct G-code machine control, stress-testing serial communication and specifying fixes for toolpath streaming failures. Compatible with **4+** industry standard printer models.
- Scripted custom toolpaths that bypass traditional CAD/CAM software in 3D printers, tuning low-level machine parameters such as extrusion rate and bed temperature to reach geometries standard slicers could not produce.
- Produced tactile graphics for visually impaired learners through programmatic texture generation and non-planar 3D prints.

Firmware Developer Oct. 2024 – June 2025
Washington Superbike | *C++, Arduino, Raspberry Pi Pico, Teensy, CAN Bus* *Seattle, WA*

- Designed a Hardware-in-the-Loop (HIL) system using Arduino, Raspberry Pi Pico, and Teensy with C++ to validate motor controller performance in electric motorcycles.
- Implemented CAN bus communication to simulate fault scenarios including battery overheating and precharge circuit failures, reducing undetected fault conditions by **∼70%** prior to physical motor controller testing.

Software Engineering Intern Aug. 2023 – July 2024
Fair Worlds | *JavaScript, HTML, 8th Wall.js, Adobe Aero* *Seattle, WA*

- Engineered geospatial web AR experiences using 8th Wall.js, JavaScript, and HTML for a VR/AR company specializing in immersive projects for major clients such as Amazon, AMD, and Dell.
- Co-developed the Seattle Art Tour in Niantic 8th Wall by 3D scanning **50+** public installations, animating assets in Adobe Aero, and implementing image recognition for AR navigation.

Projects

The Mirrors | *Unity, C#, Adobe Suite* December 2025 – March 2026

- Co-developed a horror-mystery game in Unity built around an enemy visible only through mirror reflections, requiring reflected geometry to stay consistent with the player's real-time viewpoint.
- Engineered a custom ray tracing system in C# to compute mirror reflections, correcting perspective distortion caused by off-axis viewing angles and non-coplanar mirror surfaces.
- Solely produced all 2D art, character animations, and environmental assets.

Technical Skills

Languages: Java, C#, JavaScript, Python, C/C++, HTML/CSS, G-code, Assembly
Developer Tools: Unity, Git, VS Code, Visual Studio, IntelliJ, Arduino, Blender, Adobe Suite, Onshape